X-ray:

X-rays were accidently discovered by Wilhelm Röntgen in 1895. They are electromagnetic waves of short wavelengths in the range 0.5 A° to 10 A°. The longer wavelength end of the spectrum is known as soft x-rays and the shorter wavelength end is known as hard x-ray. X-rays can be produced by several methods and has wide applications.

Production:

The Coolidge Tube: X-rays are produced when fast moving electrons are suddenly stopped by a solid target. The coolidge tube consist of a vacuum tube with a pressure of 10^{-5} mm of Hg. There are two electrodes cathod (C) and anode (A) at the two sides of the tube. There is a tungsten filament (F) heated by low tension battery/source. Cathode & Anodes are connected to a high tension supply. There is a cooling system to remove the heat generated during the production of X-ray.

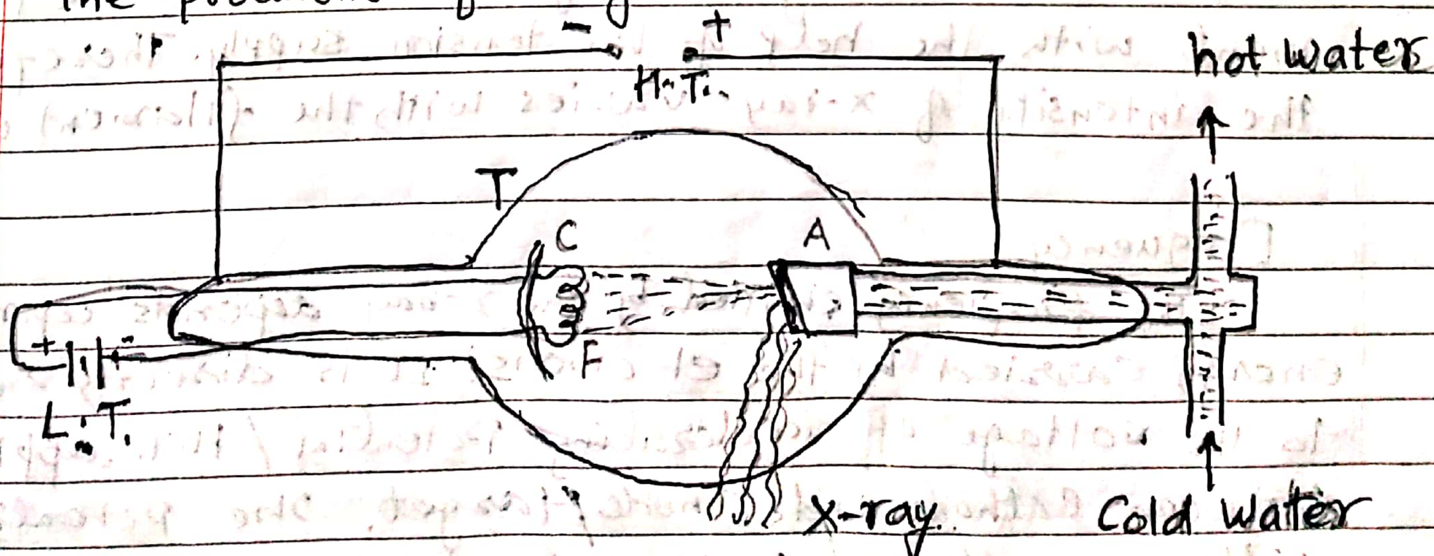

fig. production of x-ray by coolidge tube.

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The thermally heated filament is able to produced excited electrons on its surface. The thermionic electrons emitted by the filaments are accelerated towards the target/anode by the application of high voltage/H.T. supply. The target is made of a metal like tungsten or molybdenum having high melting point and high atomic number. Such type of metal target can produce more energetic and intense x-rays. The physical condition of the experiment can produce x-rays of different frequency as well as intensity. The intensity & frequency can be controlled as follows.

Intensity

The number of electrons striking the target per second determines the intensity of x-ray. Higher the temperature of filament more the number of electrons. So number of electrons emitted by the filament is proportional to its temperature. It can be adjusted by varying the current in the filament with the help of low tension supply. Therefore the intensity of x-rays varies with the filament current.

Frequency:

The frequency emitted by the x-ray depends upon the energy carried by the electrons. It is directly related to the voltage of accelerating potential/H.T. applied between cathode and anode/target. The potential difference applied be V. and charge of an electron be e. The work done on the electron in moving

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from the Cathode to the target = eV. The electron gains kinetic energy which further is converted into radiant energy called X-rays, on stricking the target. Suppose whole of the kinetic energy be converted into X-ray production, so maximum frequency of X-ray produced by ν_{max}. So, in equilibrium,

$$eV = h\nu_{max}.$$

$$\Rightarrow \nu_{max} = \frac{c}{\lambda_{min}} \qquad (\because c = \nu\lambda)$$

Thus minimum wavelength of the X-ray

$$\lambda_{min} = \frac{c}{\nu_{max}} = \frac{c}{eV/h} = \frac{hc}{eV}$$

$$\text{or} \quad \lambda_{min} = \frac{hc}{e} \cdot \frac{1}{V}.$$

$$\text{or} \quad \lambda_{min} = \frac{6.625\times10^{-34} \times 3\times10^{8}}{1.602\times10^{-19}} \cdot \frac{1}{V} = \frac{1.24\times10^{-8}}{V}\ m.$$

$$\text{or} \quad \lambda_{min} = \frac{12400}{V}\ A^{o} \qquad (1\ A^{o} = 10^{-10}\ m).$$

$$\Rightarrow \lambda_{min} \propto \frac{1}{V}.$$

The relation shows that higher the value of accelerating potential shorter will be the wavelength of X-ray emitted or greater will be the frequency. (Since $\nu_{max} \propto \frac{1}{\lambda_{min}}$)

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Types of x-ray:

Basically there are two kinds of X-rays on the basis of their origin. They are (i) Continuous x-ray
(ii) Characteristics or line X-ray.

(i) Continuous x-ray:

While bombarding high velocity electrons from the cathode to the anode, some of them penetrate the interior of the atoms of the target material and are attracted by the positive charge of the nuclei. As an electron passes close to the positive nucleus, it is deflected from its path. The electron experiences deacceleration during its deflection in the strong field of the nucleus. The energy lost during this deacceleration is given off in the form of x-radiation of continuously varying wavelength. The x-rays consist of continuous range of frequency upto a maximum frequency. This spectrum is known as continuous x-ray. This spectrum has

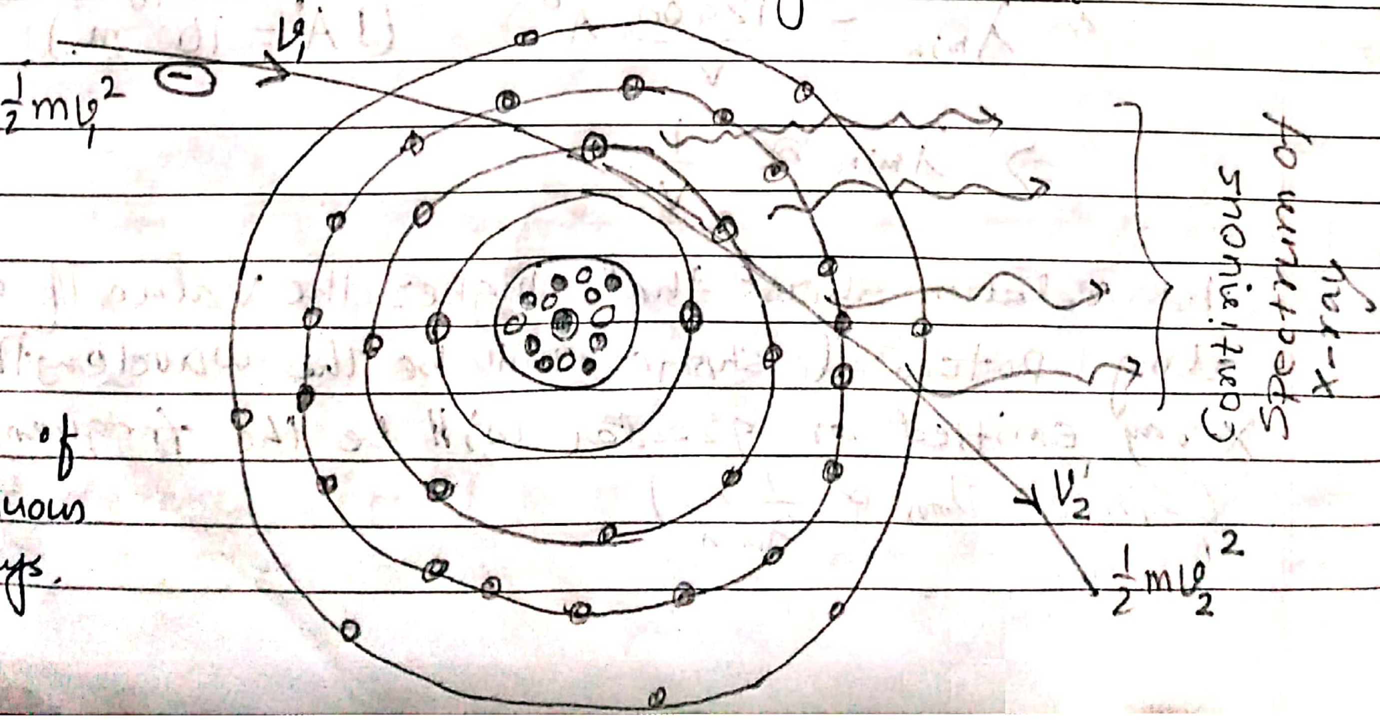

fig. Origin of continuous x-rays.

a sharply defined short wavelength limit λ_{min} (or high frequency ν_{max}), corresponding to the maximum energy of the incident electron.

Suppose striking electron has its velocity reduced from v_1 to v_2 during its passage through the atom of the target material. Then its loss of kinetic energy

$$K.E = \frac{1}{2} m (v_1^2 - v_2^2).$$

This energy must equal to the energy of X-ray photon emitted $= h\nu$.

$$\therefore \frac{1}{2} m (v_1^2 - v_2^2) = h\nu.$$

The maximum frequency of the emitted X-rays corresponds to the case when the electron ~~so~~ is completely stopped i.e. $v_2 = 0$.

So, $\frac{1}{2} m v^2 = h\nu_{max}$. ---(1) ($v_1 = v$ say, $v_2 = 0$)

If the electron is accelerated through a potential of V volts, then electrical energy eV should balance kinetic energy of the electron.

$$\therefore \frac{1}{2} m v^2 = eV. \quad \text{---}(2)$$

From equation (1) and (2), we get

$$h\nu_{max} = eV.$$

$$\text{or } \nu_{max} = \frac{eV}{h}$$

$$\text{or } \frac{c}{\lambda_{min}} = \frac{eV}{h} \qquad (\because c = \nu\lambda)$$

$$\text{or } \boxed{\lambda_{min} = \frac{hc}{eV}} \qquad \boxed{\lambda_{min} = \frac{12400}{V} A^\circ}$$

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Such type of x-rays are very aptly called braking radiation. It is because, they are produced due to braking or slowing down of high velocity electrons in the positive field of nucleus. These radiations constitute the continuous spectrum of the x-rays, because they consists of series of uninterrupted wavelengths having a sharp defined short wavelength limit λ_{min}. These x-rays are independent of the nature of the target material but are determined by the potential difference between the cathode and anode of the x-ray tube.
(Bremsstrahlung radiation = Braking radiation).

(ii). Characteristic or Line X-ray.

The electrons in X-ray tube, some of the high velocity electrons while penetrating the interior of the atoms of the target material, knock off the tightly bound electrons in the innermost shells (K, L - etc) of the atom. When electron from outer orbits jump to fill up the vacancy so produced, the energy difference is given out in the form of x-rays of definite wavelength/frequency. These wavelengths constitute the line spectrum which is characteristic of the material of the target.

When an electron from K-shell is knocked a vacancy is created in it. It may be filled by a nearby electron in the L-shell. During this jump an x-ray radiation is emitted, whose frequency

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Can be expressed as

$$E_K - E_l = h\nu.$$

where E_K is the energy required to dislodge an electron from the K-shell & E_L is that required for L-shell. These X-rays are highly penetrating.

If, however, the vacancy in K-shell is filled up by an electron jumping from M-shell, the X-rays emitted would be still more energetic and would consequently possesses still higher frequency ν'.

So, $$E_K - E_m = h\nu'$$

Similarly, when the incident electron carries lesser amount of energy, it dislodges an electron from L-orbit or from M-orbit or other outer orbits. In this case X-rays of lower frequency than that of K-series are produced.

In fig electron 1 with high energy knocks electron 2 from K-orbit. The vacant position in K-orbit is filled up by electron 3 from L-orbit. During transition of electron from L-orbit to K-orbit, X-ray is produced.

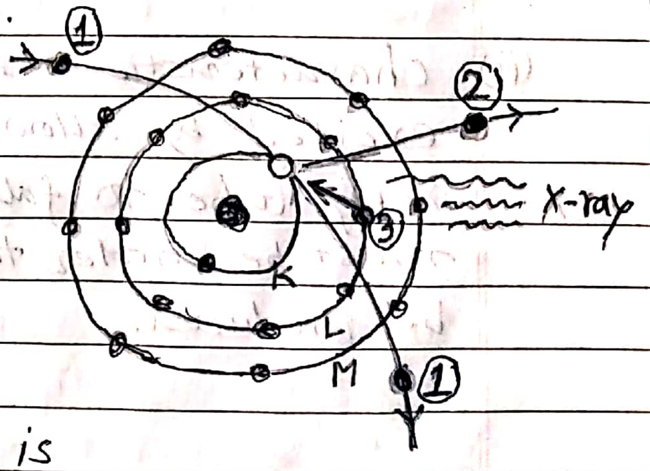

fig. production of characteristic X-ray

In this way X-ray spectra of K-series, L-series, M-series .. etc are produced.

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when transition of electron ends at K-orbit, L-orbit, M-orbit .. and so on. There is possibility of transition of electron from any higher orbit electron to K-orbit, forming X-ray of K-Series. Likewise X-ray of L-series, M-series etc are produced when jumping of electrons from any higher orbit to L-orbit, M-orbit etc takes place.

There are two methods of producing characteristic X-rays.

(i), The characteristic X-rays of an element can be excited by using the element as the target in the X-ray tube & thus subjecting it to direct bombardment of electrons. For each target there is a minimum potential below which the line spectra do not apper, which is different for different target. e.g. for Molybdenum 35 KV and Tungsten 70 KV.

(ii). Characteristic X-rays of an element can also be excited by allowing primary X-rays from a hard X-ray tube to fall on the element. The primary X-rays must be harder than the characteristic X-rays to be produced.

Origin of characteristic X-rays:

The origin of characteristic X-rays can be understood in terms of Bohr's theory. Suppose an

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atom in the target of an x-ray tube is bombarded by high speed electron and a K-electron is removed. A vacancy is created in the K-shell. This vacancy can be filled up by an electron from either L, M or N shells or a free electron. When vacant region in K-shell is filled by L-electron, it produces K_α line and when filled by M-electron or N-electron, it produces K_β or K_γ lines respectively. The range of wavelength of K-series ~~is~~ lies 0.486 A° to 0.563 A°. Similarly, the longer wavelength L-series originates when an L-electron is knocked out of the atom. Thus L_α, L_β ... lines are produced, whose wavelength range lies in 3.3 A° to 4.7 A°. Likewise M-series can also be observed when M-electron is knocked out and so on.

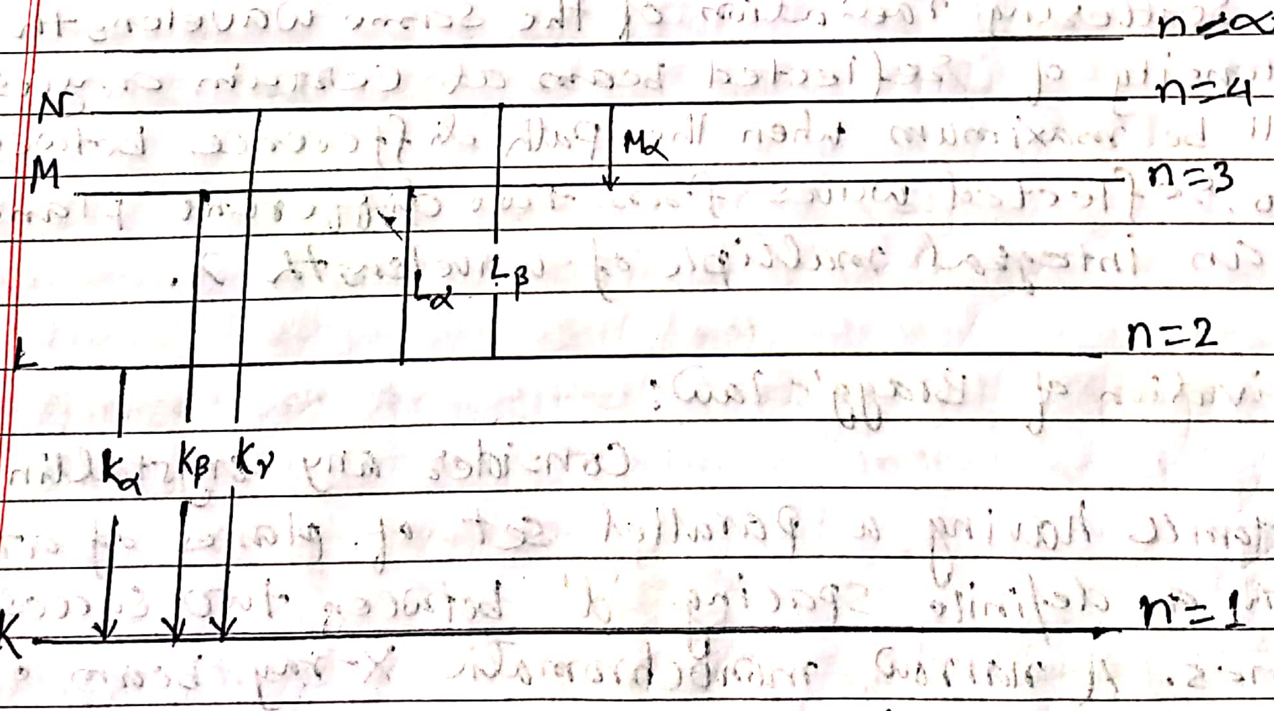


fig. characteristic x-rays of different series

Further K-series is composed of four lines as $K_{\alpha 1}$, $K_{\alpha 2}$, $K_{\beta 1}$, $K_{\beta 2}$. Likewise L-series is composed of more than 30 lines and so on. They are designated as $L_{\alpha 1}$, $L_{\alpha 2}$, $L_{\beta 1}$, $L_{\beta 2}$ and so on. These lines are known as fine structure.

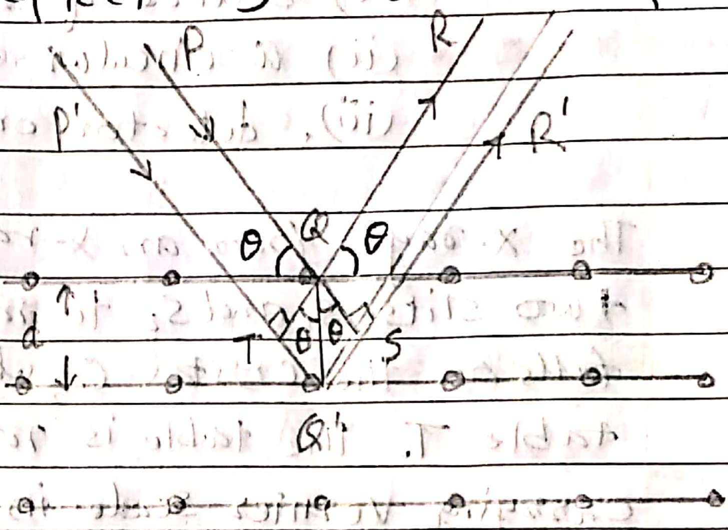
Bragg's law:

The reflection of x-rays is also according to common laws of reflection of light. When monochromatic x-rays impinge upon the atoms in a crystal lattice, each atom acts as a source of scattering radiation of the same wavelength. The intensity of reflected beam at certain angles will be maximum when the path difference between two reflected waves from two different planes is an integral multiple of wavelength λ.

Derivation of Bragg's law:

Consider any crystalline substance having a parallel set of planes of atoms with a definite spacing 'd' between two successive planes. A narrow monochromatic X-ray beam of wavelength λ be falling on the first plane at a glacing angle θ. The ray PQ be incident on the first plane. The corresponding reflected ray QR

Should be reflected with same angle θ. Since X-rays are much more penetrating than ordinary light, there is only partial reflection at each plane. Consider two parallel rays PQR and P'Q'R' in the x-ray beam which are reflected by two atoms Q & Q', as shown in fig. The atom Q' is vertically below Q. The ray P'Q'R' has a longer path than ray PQR. To compute the path difference between the two rays, from Q, draw normal QT and QS on P'Q' and Q'R' respectively. Then the path difference = TQ' + Q'S. From two right angled triangles QTQ' and QSQ', we get $\sin\theta = \frac{TQ'}{QQ'} = \frac{SQ'}{QQ'}$



$$\text{So } TQ' + Q'S = QQ'\sin\theta + QQ'\sin\theta$$
$$= d\sin\theta + d\sin\theta$$
$$= 2d\sin\theta.$$

Hence two reflected X-rays will reinforce each other and produce maximum intensity if satisfies the following condition

$$2d\sin\theta = n\lambda.$$

where $n = 1, 2, 3 \ldots$ order of the scattered beam. and λ is the wavelength of X-ray. This equation is called Bragg's law.

The Bragg's X-ray spectrometer :-

Construction

The spectrometer is similar to an optical spectromoter. It consists of three parts
(i) source of x-rays
(ii) a circular table with crystal
(iii). detector or ionisation chamber.

The x-rays from an x-ray tube are passed through two slits S_1 and S_2 to make a fine beam. The beam falls on the crystal C, which is mounted on circular table T. The table is rotable about a vertical axis, carrying vernier scale in circular fashion. The table is provided with a radial arm (R) carrying an ionisation chamber (I). It is also rotable about the same vertical axis. The ionisation chamber is connected to an electrometer (E) to measure ionisation current. The intensity of reflected/diffracted x-ray can be measured which enters into ionisation chamber. There is a third slit S_3 inside the chamber to colimate the beam. The arrangement works on the principle that, when the crystal rotates through an angle θ, the ionisation chamber turns through angle 2θ.

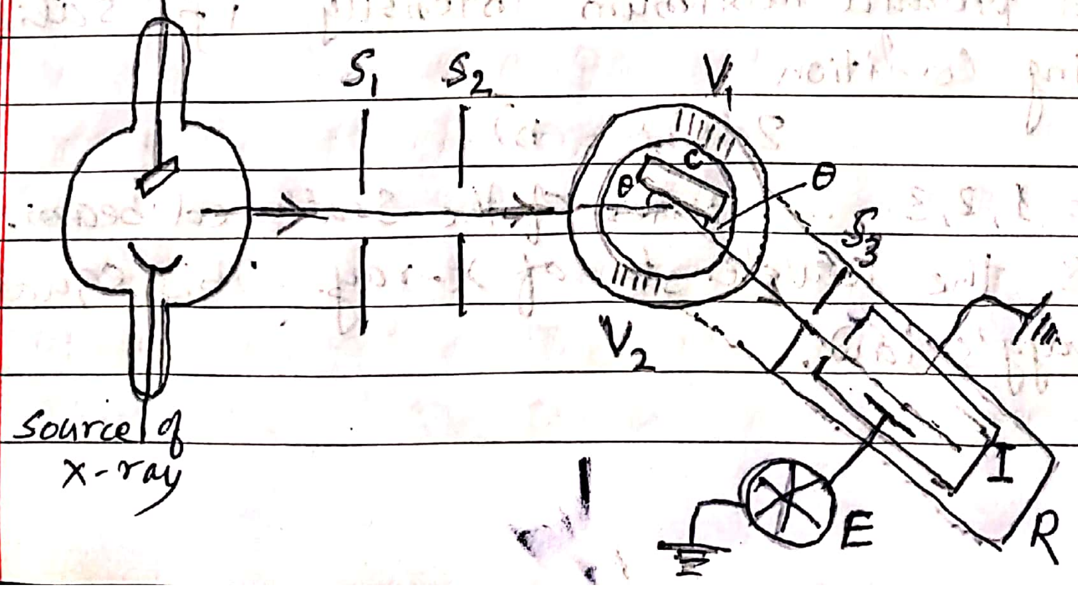


fig. Bragg's X-ray spectrometer.

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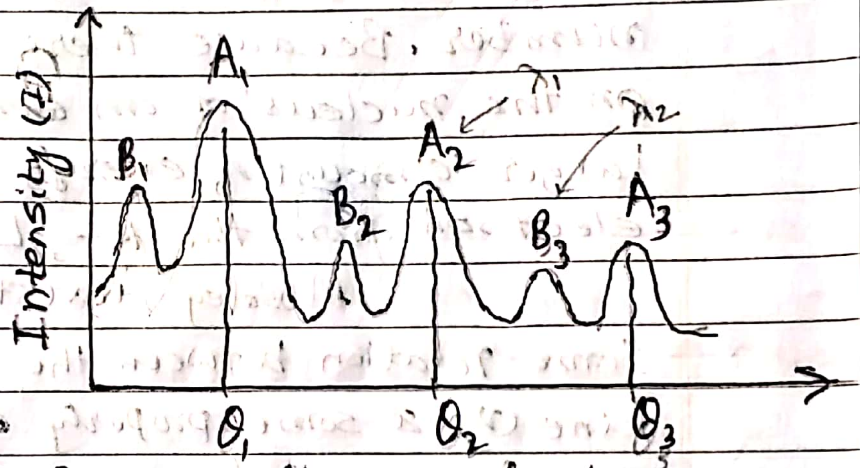
Working

Initially, the crystal is placed for small glacing angle θ, It is then increased gradually in small step and corresponding ionisation current is noted. The amount of current is proportional to the intensity of x-ray diffracted. A graph of ionisation current against the glancing angle is obtained as shown in fig. It is called an x-ray spectrum.

The prominent peaks A_1, A_2, A_3.. etc refer to x-rays of wavelength λ. The glancing angles $\theta_1, \theta_2, \theta_3$.. corresponding to the peaks A_1, A_2, A_3.



It is found that
$\sin\theta_1 : \sin\theta_2 : \sin\theta_3 = 1:2:3$.
The peaks A_1, A_2, A_3 etc refer to the first, second and third order reflection of the same wavelength λ. Similarly B_1, B_2, B_3 peaks are for same order in case of another wavelength λ_2.

Moseley's Law:—

In 1913-14, Moseley carried out a systematic study of the characteristic X-ray spectra of various elements used as targets in an X-ray tube. By using Bragg's spectrometer for the purpose, he found that the characteristic X-ray spectra of different elements are remarkably similar to each other in the sense that each consists of K-, L-, M- series. However there is one very important difference. The frequency of lines produced from an element of higher atomic number is greater than that produced by an element of lower atomic number. It is due to the fact that binding energy of electrons increases as we go from one element to another of higher of higher atomic number. Because there is greater positive charge on the nucleus of an element of higher atomic number, larger amount of energy is required to liberate an electron from the K-, L- and M-shells of that element.

Moseley investigated the possibility of some relation between the frequency of a given spectral line (ν) & some property of the atom emitting the line. He first plotted square root of frequency ($\sqrt{\nu}$) with atomic weight (A). The frequency did not vary uniformly with atomic weight. Then he plotted square root of frequency ($\sqrt{\nu}$) of a given line (say K_α) against the atomic number (Z) of the elements emitting that line. He obtained a straight line as shown in fig. The same linear relation was found to hold good for any line in any series. He concluded, therefore, that atomic number (not the atomic weight) is the fundamental property of elements.

Statement of Moseley's law:

The frequency of a spectral line, in X-ray spectrum, varies as the square of the atomic number of the element emitting it, which can be expressed as

$$\nu \propto Z^2.$$

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Mosley's law can also be written as

$$\sqrt{\nu} = a(Z-b)$$

Here Z = atomic number of the element

a, b = Constants depending upon the particular line.

The values of constants a and b vary from one series to another. The values in K-series are different from those of L-series. The constant 'a' is proportionality constant and 'b' is nuclear screening constant

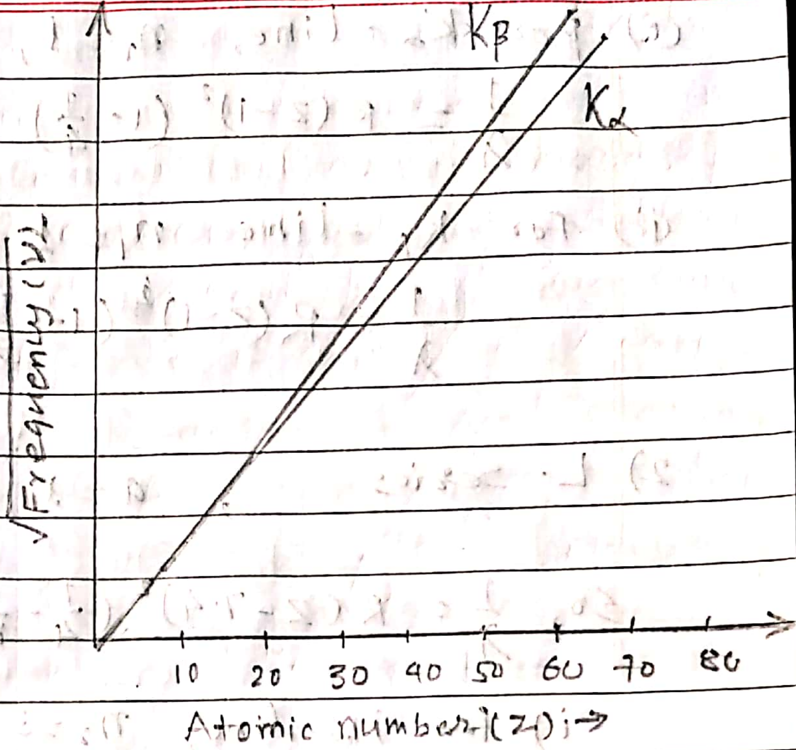

fig: The plot of $\sqrt{\nu}$ Vs Z of various lines of X-ray lines.

Explanation according to Bohr's theory:

Bohr's theory of hydrogen spectrum gives the frequency of a spectral line as

$$\nu = Z^2 Rc\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

where R = Rydberg's constant & c = Velocity of light.

The frequencies for various lines in the K- and L-series of the X-ray line spectrum are expressed as -

1) K- series:

The general formula $\nu = Z^2 Rc\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

or $\frac{1}{\lambda} = R(Z-1)^2\left(\frac{1}{1^2} - \frac{1}{n_2^2}\right)$

Taking a = 1 & b = 1.

where $n_1 = 1$, $n_2 = 2, 3 \ldots$

(i) For K_α-line, $n_1 = 1$, $n_2 = 2$

$$\frac{1}{\lambda} = R(Z-1)^2\left(1-\frac{1}{4}\right) = \frac{3}{4}R(Z-1)^2.$$

(ii) For K_β-line $n_1 = 1$, $n_2 = 3$

$$\frac{1}{\lambda} = R(Z-1)^2\left(1-\frac{1}{9}\right) = \frac{8}{9}R(Z-1)^2.$$

2) L-Series, $a = 1$, $b = 7.4$, $n_1 = 2$ & $n_2 = 3, 4 \ldots$

So, $$\frac{1}{\lambda} = R(Z-7.4)^2\left(\frac{1}{4}-\frac{1}{n_2^2}\right)$$

i) For H_α-line, $n_2 = 3$

$$\frac{1}{\lambda} = R(Z-7.4)^2\left(\frac{1}{4}-\frac{1}{9}\right) = \frac{5}{36}R(Z-7.4)^2$$

and so on.

Importance of Moseley's law:

1) According to this law, it is the atomic number and not atomic weight of an element which determines its characteristic properties, both physical & chemical. Hence elements are arranged in periodic table on the basis of atomic number. For example $_{18}Ar^{40}$ (Argon) comes before $_{19}K^{39}$ (Potassium), $_{27}Co^{58.9}$ (cobalt) comes before $_{28}Ni^{58.7}$ (Nickel) etc.

2) Mosoley's work has also helped to perfect the periodic table by discovery of new elements e.g. Masurium(43), Rhenium (75) etc

3) It also helps to determine atomic numbers of rare-earth elements & fixing their position in the periodic table.

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Compton Scattering:

The Compton Effect: Compton discovered that when x-rays of a sharply defined frequency were incident on a material of low atomic number like carbon, they ~~β~~ suffered a change of frequency on scattering. The scatted beam contains two wavelengths. In addition ~~to the spectral~~ there exists a line of longer wavelength. The change of wavelength is due to the loss of energy of the incident x-rays. This elastic interaction is known as Compton effect.

This effect was explained by Compton on the basis of quantum theory of radiation. In this process both energy & momentum are conserved for both colliding agents, of photon & electron. During collision energy of photon is transferred to the electron. As a result decrease in frequency takes place.

Suppose an incident photon with an energy $h\nu$ and momentum $h\nu/c$ strikes an electron at rest. The initial momentum of the electron is zero and energy is rest mass energy, m_0c^2. The scattered photon of energy

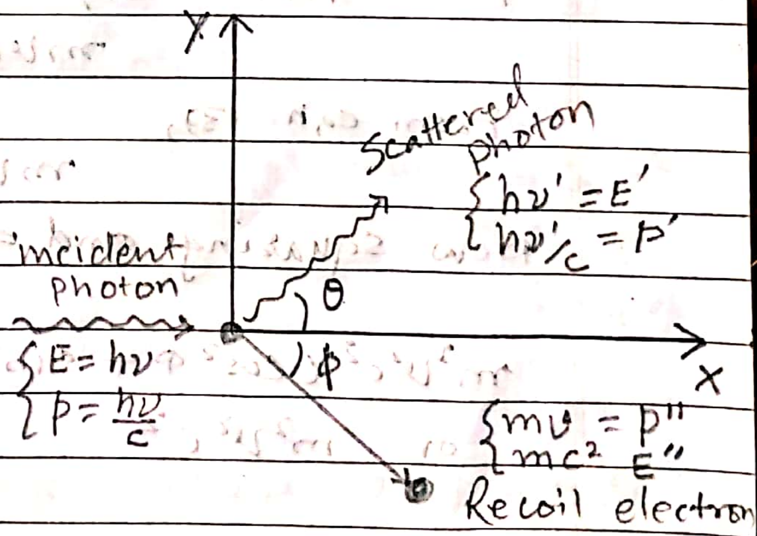

fig. Compton scattering

$h\nu'$ and momentum $h\nu'/c$ moves off in a direction inclined at an angle θ to the original direction. The electron acquires a momentum mv and moves at an angle ϕ to the original direction. The energy of recoil electron is mc^2.

According to the principle of conservation of energy

$$h\nu + m_0c^2 = h\nu' + mc^2 \quad \text{---(1)}$$

Again considering the principle of conservation of momentum, along x-direction,

$$\frac{h\nu}{c} = \frac{h\nu'}{c}\cos\theta + mv\cos\phi. \quad \text{---(2)}$$

and in y-direction

$$0 = \frac{h\nu'}{c}\sin\theta - mv\sin\phi. \quad \text{---(3)}$$

From equation (2),

$$mvc\cos\phi = h(\nu - \nu'\cos\theta) \quad \text{---(4)}$$

From eqn (3),

$$mvc\sin\phi = h\nu'\sin\theta \quad \text{---(5)}$$

Now squaring and adding eqn (4) & (5),

$$m^2v^2c^2(\cos^2\phi + \sin^2\phi) = \{h(\nu - \nu'\cos\theta)\}^2 + (h\nu'\sin\theta)^2$$

$$\text{or } m^2v^2c^2 = h^2(\nu^2 - 2\nu\nu'\cos\theta + \nu'^2\cos^2\theta + h^2\nu'^2\sin^2\theta;$$

$$= h^2(\nu^2 - 2\nu\nu'\cos\theta) + h^2\nu'^2(\cos^2\theta + \sin^2\theta)$$

$$= h^2(\nu^2 - 2\nu\nu'\cos\theta + \nu'^2)$$

$\therefore$

$\therefore\ m^2 v^2 c^2 = h^2(\nu^2 - 2\nu\nu'\cos\theta + \nu'^2)$ --- (6).

Also from eqn (i),

$$mc^2 = h(\nu - \nu') + m_0 c^2$$

squaring both sides,

$$m^2c^4 = h^2(\nu^2 - 2\nu\nu' + \nu'^2) + 2h(\nu-\nu')m_0c^2 + m_0^2c^4. \quad \text{--- (7)}.$$

Subtracting equation (6) from (7), we get,

$$m^2c^2(c^2 - v^2) = h^2\nu^2 - 2h^2\nu\nu' + h^2\nu'^2 + 2h(\nu-\nu')m_0c^2 + m_0^2c^4 - h^2\nu^2 + 2h^2\nu\nu'\cos\theta - h^2\nu'^2.$$

$$= -2h^2\nu\nu'(1-\cos\theta) + 2h(\nu-\nu')m_0c^2 + m_0^2c^4 \quad \text{---(8)}.$$

From relativistic formula, for the electron,

$$m = \frac{m_0}{\sqrt{1 - v^2/c^2}}$$

On squaring, $m^2 = \dfrac{m_0^2}{1 - v^2/c^2} = \dfrac{m_0^2c^2}{c^2 - v^2}$

or, $m^2c^2(c^2 - v^2) = m_0^2c^4$ --(9) (multiplying by c^2 on both sides).

From eqn (8) & (9), [rearranging eqn (8) with the help of eqn. (9)]

$$\cancel{m_0^2c^4} = -2h^2\nu\nu'(1-\cos\theta) + 2h(\nu-\nu')m_0c^2 + \cancel{m_0^2c^4}.$$

$$2h(\nu-\nu')m_0c^2 = 2h^2\nu\nu'(1-\cos\theta)$$

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or, $$\frac{\nu-\nu'}{\nu\nu'} = \frac{h}{m_0c^2}(1-\cos\theta)$$

or, $$\frac{1}{\nu'} - \frac{1}{\nu} = \frac{h}{m_0c^2}(1-\cos\theta).$$

or, $$\frac{c}{\nu'} - \frac{c}{\nu} = \frac{h}{m_0c}(1-\cos\theta).$$

or, $$\lambda' - \lambda = \frac{h}{m_0c}(1-\cos\theta). \quad \text{---} (10)$$

$\therefore$ The change in wavelength $= d\lambda = \frac{h}{m_0c}(1-\cos\theta)$

Above relation shows that change in wavelength is independent of the wavelength of the incident radiation as well as the nature of the scattering substance. It depends only on the angle of scattering θ.

Following may be the cases:

I) when $\theta = 0$, $\cos\theta = 1$, Hence $d\lambda = 0$
i.e. there is no any change in wavelength of radiation if photon is not scattered.

II) when $\theta = 90°$, $\cos\theta = 0$

$$\Rightarrow d\lambda = \frac{h}{m_0c} = \frac{6.63\times10^{-34}}{9.11\times10^{-31}\times3\times10^{8}} = 0.0243 A°.$$

This is known as Compton wavelength.

III) when $\theta = 180°$, $\cos\theta = -1$, $d\lambda = \frac{2h}{m_0c} = 0.0485 A°$
which is the maximum value.

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